rbGyanX — Plan Evaluation Report

One Patient · One Plan · Complete Evaluation

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Patient / Plan

Patient ID: RBX-TXT-001
Plan: RBX-TXT-001
Site: HN · Technique: IMRT
Fractionation: 66 Gy / 33 fx

Structure

PTV 66 (target) · Literature organ: PTV · Composite plan (4 structures)

Radiobiology (Poisson-LQ (DVH))

TCP 100.0% (base 100.0% → covariate-adjusted 100.0%) · BED 79.20 Gy · EQD2 66.00 Gy

Clinical data: PTV extension (synthetic-flagged) (synthetic-flagged)

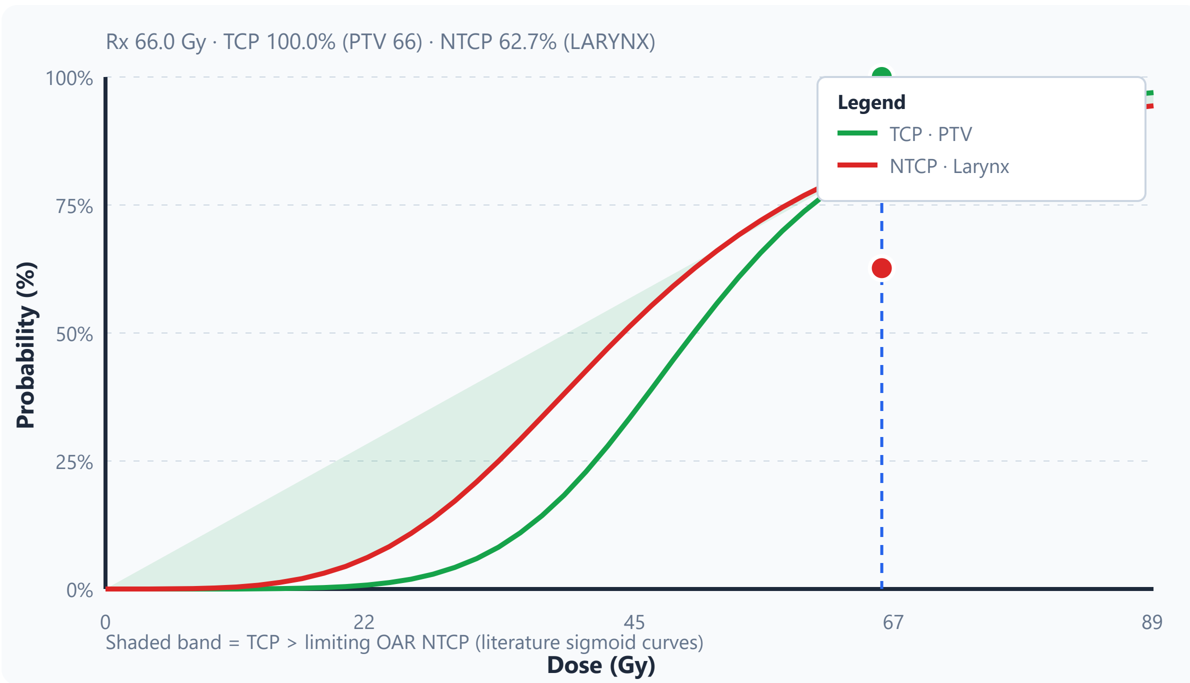
Mean 67.31 Gy · Max 71.90 Gy · gEUD 67.49 Gy

Dose metrics (QUANTEC-oriented)

Metric	Value	Note
D98% (near-min)	64.3 Gy	RTOG/ICRU coverage
D95%	65.4 Gy	RTOG/ICRU coverage
D50% (median)	67.6 Gy	ICRU reference
D2% (near-max)	69.4 Gy	Hot spot (serial OAR proxy if needed)
Dmean	67.3 Gy	Mean dose

Dmax	71.9 Gy	Maximum dose
V95% Rx	99.9%	Volume ≥ 95% prescription
V100% Rx	92.3%	Volume ≥ prescription
V107% Rx	0.0%	Hot volume above Rx

Therapeutic window — TCP and NTCP dose–response at plan prescription



Composite plan — all structures

Primary target: PTV 66

Structure	Type	Organ	Model	TCP/NTCP	Dmean	Dmax	D95
PTV 66	target	PTV	poisson_dvh	TCP 95.0% (model 100.0%)	67.31 Gy	71.90 Gy	65.4 Gy
LARYNX	oar	Larynx	lkb_loglogit	NTCP 62.7%	49.91 Gy	70.20 Gy	38.8 Gy
Parotid	oar	Parotid	lkb_loglogit	NTCP 53.8%	29.35 Gy	59.80 Gy	8.7 Gy

Spinal Cord	oar	Spinal Cord	lkb_loglogit	NTCP 0.0%	27.73 Gy	42.54 Gy	1.5 Gy

PTV 66 — physical indices

Metric	Value	Note
D98% (near-min)	64.3 Gy	RTOG/ICRU coverage
D95%	65.4 Gy	RTOG/ICRU coverage
D50% (median)	—	ICRU reference
D2% (near-max)	69.4 Gy	Hot spot (serial OAR proxy if needed)
Dmean	67.3 Gy	Mean dose
Dmax	71.9 Gy	Maximum dose

LARYNX — physical indices

Metric	Value	Note
D2% (near-max)	68.1 Gy	QUANTEC serial OAR
D0.03 cc	70.2 Gy	Max point dose (approx.)
Dmax	70.2 Gy	Maximum dose
gEUD (a=1)	50.1 Gy	Used in NTCP models

Parotid — physical indices

Metric	Value	Note
Dmean	29.4 Gy	QUANTEC parallel OAR (primary)
Dmax	59.8 Gy	Maximum dose
V30 Gy	—	QUANTEC xerostomia (V30)
V25 Gy	—	Salivary sparing

gEUD (a=1)	29.5 Gy	Used in NTCP models

Spinal Cord — physical indices

Metric	Value	Note
D2% (near-max)	41.3 Gy	QUANTEC serial OAR
D0.03 cc	42.5 Gy	Max point dose (approx.)
Dmax	42.5 Gy	Maximum dose
gEUD (a=1)	27.8 Gy	Used in NTCP models

Target plan indices

Index	Value	Note
TCI (% vol ≥ Rx)	92.3%	Target coverage
CI (RTOG)	0.923	Conformity
HI (ICRU)	1.079	D2%/D98%
HI (modified)	0.075	(D2−D98)/D50
D98%	64.3 Gy	
D95%	65.4 Gy	
D2%	69.4 Gy	
V95% Rx	99.9%	
V100% Rx	92.3%	

Therapeutic window

- TCP 100.0% (covariate-adjusted from base 100.0%)
- Composite NTCP 62.7%
- Critical OAR NTCP 0.0%
- UTCP 16.4%

- P+ 95.0%
- TWI 35.0 (Favorable)

Clinical context (opt-in)

Cancer site: Head & Neck

Clinical covariates were applied to TCP/NTCP using exploratory log-odds adjustment (age, sex, chemo, smoking, ECOG, organ-specific dose slopes). Base DVH-derived probabilities are shown in parentheses where adjusted values are displayed. Synthetic-flagged rows remain unsuitable for toxicity correlation.

Patient

Field	Value
Age (years)	55
Sex	Male

Disease & site

Field	Value
Histology	Head and Neck Cancer
T stage	Unknown
N stage	Unknown
Smoking status	Unknown

Treatment

Field	Value
Concurrent chemotherapy	Yes

Organ guideline

ICRU Reports 50/62/83 — target volume definitions and dose reporting.

References

- Bentzen SM, et al. QUANTEC: An introduction to the scientific issues. Int J Radiat Oncol Biol Phys. 2010;76(3 Suppl):S3-S9.
- Marks LB, et al. Use of normal tissue complication probability models in the clinic. Int J Radiat Oncol Biol Phys. 2010;76(3 Suppl):S10-S19.
- Zaider M, Minerbo G. Tumour control probability for a uniform stage III lung tumour: clinical implications for dosimetric studies. Phys Med Biol. 2000;45(1):199-213.